

PROJECT

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MULTI-CLOUD DATA TRANSFER WITH AWS AND GCP

MULTI-CLOUD SERVICE

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Objective

This project aims to demonstrate how to securely transfer data from AWS S3 to Google Cloud Storage using GCP's Storage Transfer Service.

It involves setting up cross-platform permissions and configuring automated object migration between cloud providers.

Services

Amazon S3 – Stores the source data in buckets on AWS.

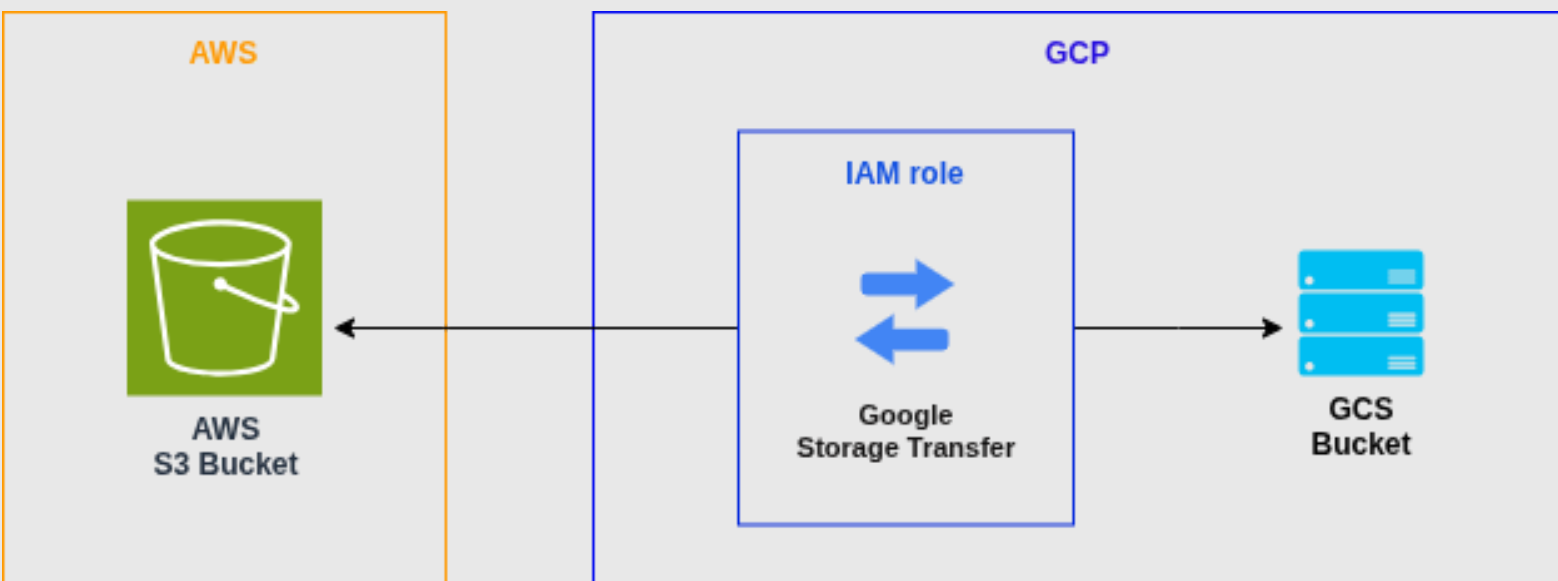
AWS IAM - Provides secure access by creating roles and permissions for data transfer.

GCP Cloud Storage (GCS) - Acts as the destination for storing transferred data in Google Cloud.

Storage Transfer Service - A Google-managed service that pulls objects from AWS S3 into GCS buckets.

Amazon SQS - Optionally used to notify GCP about object changes for incremental transfers.

Architecture Diagram



Data is stored in an S3 bucket on AWS. GCP's Storage Transfer Service connects using IAM-based temporary credentials. A GCS bucket is set as the destination, and the transfer begins once access is granted. Amazon SQS can optionally notify GCP of object changes, enabling secure and efficient multi-cloud data migration.

Steps to Implement

1. Store data in an Amazon S3 bucket:

Upload your files to a newly created or existing S3 bucket in your AWS account. Make sure the bucket is in a region supported by GCP's Storage Transfer Service.

2. Log in to AWS with an IAM user:

Create or use an IAM user with permission to access the S3 bucket. Configure the AWS CLI using `aws configure` by entering the access key, secret key, region, and output format.

3. Create a free Google Cloud account:

Sign up for GCP and activate the free trial. Set up billing and create a new project to use Google Cloud Storage and the Storage Transfer Service.

4. Create a destination GCS bucket:

In GCP, create a new Cloud Storage bucket in your chosen region to receive the data from AWS S3 during the transfer process.

5. Grant GCP access to AWS S3 bucket:

Generate a service account in GCP with Storage Transfer permissions. Add an IAM policy to your AWS bucket to allow access using the service account's credentials.

6. Set up and run the Storage Transfer Job:

In the GCP console, configure a new transfer job using the source S3 bucket, AWS credentials, and destination GCS bucket. Optionally, enable SQS for tracking object changes during recurring transfers.

Story Time

Amazon S3 (The Cloud Treasure Chest):

S3 is your golden chest in the AWS realm where you store all your shiny files, scrolls, and digital treasures, safe and ready for travel.

IAM User (The Access Guardian):

IAM is the guardian who holds the enchanted keys. They allow the chosen ones (like GCP) to peek into your treasure chest, only if you grant them the right scrolls (permissions).

GCP Account (The Second Kingdom):

GCP is a neighboring kingdom with its own cloud towers. By creating a portal (account), you can start building a path to share treasures between kingdoms.

GCS Bucket (The Magic Cauldron):

The GCS bucket is where your treasures arrive in the GCP realm. It's a magic cauldron that holds and protects everything you send from the AWS world.

Storage Transfer Service (The Teleport Wizard):

This mighty wizard from GCP casts a powerful spell to teleport objects straight from your AWS chest to the GCP cauldron, no manual carrying needed!

Amazon SQS (The Messenger Owl – Optional):

If you want to keep the wizard updated on every new scroll added to your treasure chest, the SQS owl will deliver those messages instantly for smooth, ongoing transfers.

BY
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THANK YOU!

